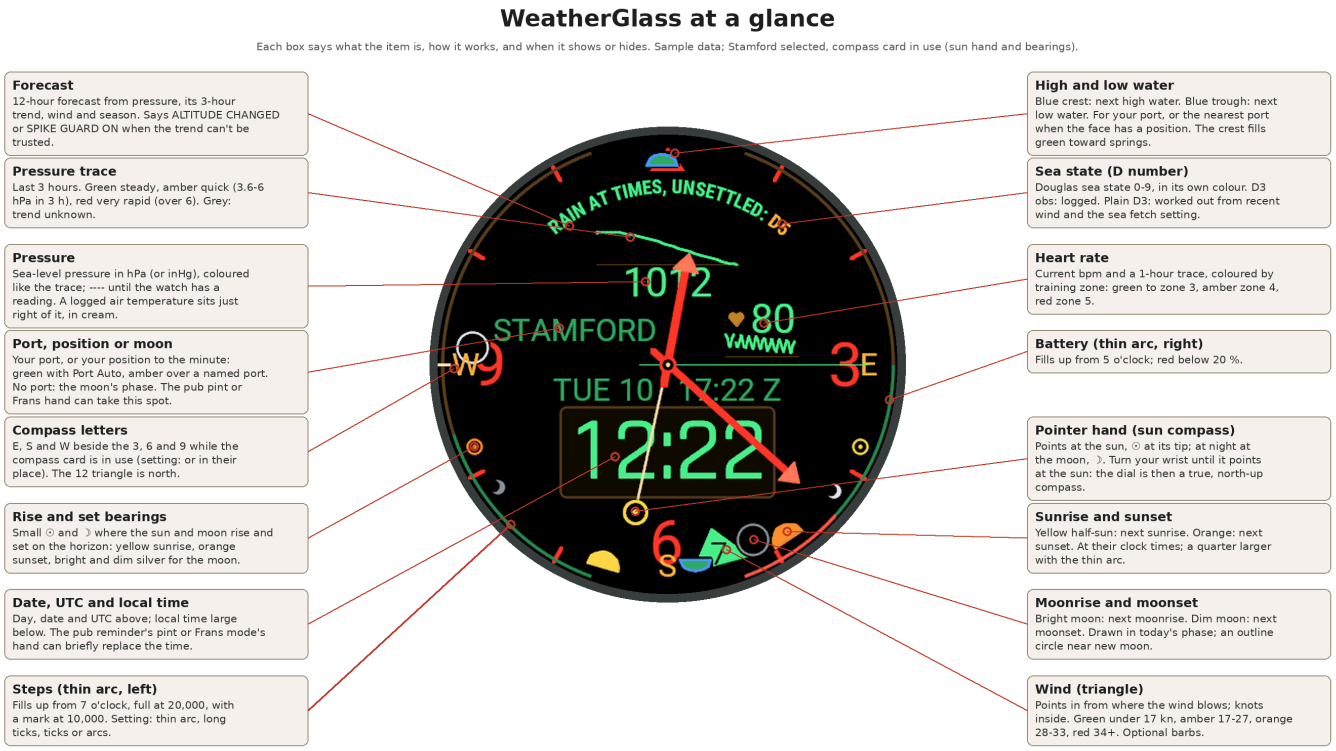
WeatherGlass — User's Guide

10 November 2026 · Larry Sherman · WeatherGlass v1.20.2, WeatherGlass Log v1.5.2



Introduction

WeatherGlass is a free Garmin watch face for sailors: barometer, forecast, wind, sun, moon and tides on one dial, worked out on the watch itself. It follows the boat: with a position, the sun, moon and tides are worked out for where you are, and a pointer hand turns the dial into a sun and moon compass. It was built for the Clipper Round the World Yacht Race, so its tide ports are the race's host ports, grouped by leg.

The design rule is simple: the important things are big, the secondary things are small. The local time and the barometer are the largest items; sun, moon, wind and tides sit as small marks around the edge of the dial. The colours are chosen to stay readable at night without the watch's red-shift mode, and an all-red scheme is there for full dark adaptation.

A weatherglass is the old seaman's word for a barometer, and the pressure trace is the face's headline feature.

A tour of the face

The weather sits at the top, the moon, your port or your position on the left, heart rate on the right, the time at the bottom, and everything else as marks on the edge of the dial.

Where	What it shows
Top, curved along the rim	The forecast, then the sea state as a Douglas number, for example FAIRLY FINE, SHOWERY LATER: D3

Where	What it shows
Top, under the forecast	3-hour pressure trace with today's sea-level pressure under it (and a logged air temperature beside it)
Left of centre	The moon's phase, your port's name, or your position in degrees and minutes (see Choosing your port)
Right of centre	Heart rate with a 1-hour trace
Bottom	Day, date and UTC on one line, with the local time large underneath
Left side of the rim	Steps: a thin green arc filling upward from 7 o'clock, full at 20,000, with a mark at 10,000
Right side of the rim	Battery: a thin green arc filling upward from 5 o'clock, red in its bottom fifth
Dial edge marks	Sunrise, sunset, moonrise, moonset, high and low water, each at its clock position, with small ☉ and ☽ symbols at the compass bearings where the sun and moon rise and set; the wind triangle sits just inside them
Pointer hand	A light cream hand with ☉ at its tip pointing at the sun (☽ and the moon at night): the sun and moon compass
Hands	Hour and minute in red, with a green seconds hand (can be turned off)

Barometer and forecast

The line at the top is the last 3 hours of pressure, oldest on the left and now on the right, with the current sea-level pressure in hPa (or inHg) underneath. Its colour tells you how fast the glass is moving, using the shipping forecast's terms for the change over 3 hours:

Colour	Change in 3 hours	Shipping forecast term
Green	under 3.6 hPa	steady, rising or falling slowly
Amber	3.6 to 6.0 hPa	rising or falling quickly
Red	more than 6.0 hPa	rising or falling very rapidly

Rises and falls are treated alike, since a rapid rise brings wind too. A rough sailor's guide: quickly often means force 6 to 8 within 6 hours; very rapidly, force 7 to 9 within 3 hours.

The forecast curved along the top is a Zambretti forecast: the classic barometer forecaster that combines pressure, its 3-hour trend, the wind direction and the season. It is a guide for the next 12 hours or so, not a replacement for an official forecast.

South of the equator the forecaster's rules change (the seasons flip and lows turn the other way). The **Hemisphere** setting handles this: Auto works it out from the length of the day, or you can set North or South by hand, which is worth doing near the equinoxes or the equator.

If the watch has no pressure reading yet, the number shows ---- and no forecast is given.

Grey barometer. Grey means the face cannot trust the 3-hour trend, so the trace and number lose their colour and the forecast line says why:

Forecast line	Why	Clears
ALTITUDE CHANGED	You climbed or descended, and the face's own pressure log does not yet cover 2½ hours	Once the log has 2½ hours of fresh readings

Forecast line	Why	Clears
SPIKE GUARD ON	The watch's own sea-level reading jumped more than any weather can, usually because the watch recalibrated its altitude	Once the jump is more than 3½ hours old

Grey means unknown, never calm: check the sky and the official forecast.

Wind

The triangle, just inside the dial edge, points inward from the direction the wind is blowing from, with 12 o'clock as north. It may cover part of the 3, 6 or 9 when the wind comes from that way. The number inside is the speed in knots.

- **Colour** shows the strength by Beaufort band: **green** F0–4 (under 17 kn), **amber** F5–6 (17–27 kn), **orange** F7 (28–33 kn), **red** F8 and above (34 kn or more).
- **Dimmer colour**: the wind report is 1 to 3 hours old.
- **Gone**: over 3 hours old, or under 1 knot. Old wind is never shown as current, and the forecast then works without it.
- **Barbs (optional)**: switch on **Wind barbs** for GRIB-style feathers along one edge of the triangle, leaning out toward the rim: half a feather 5 kn, a full feather 10, a pennant 50. As on weather charts they sit on the clockwise side in the northern hemisphere and the anticlockwise side in the southern, following the **Hemisphere** setting.

The wind comes from Garmin's weather service through your phone; it is the forecast wind for your position, not measured on the boat. Offshore without the phone, it fades out as above. If the triangle is missing ashore, opening the Garmin Connect app on the phone usually brings it back.

Sea state

The end of the forecast line, after the colon, gives the sea state on the Douglas scale, the one the shipping forecast uses: **D3** is slight, **D4** moderate, **D5** rough, **D6** very rough, **D7** high. **D3 obs** means it is your own observation from WeatherGlass Log; a plain **D3** is the face's estimate.

It has its own colour, separate from the forecast: **green** up to D4, **amber** D5, **orange** D6, **red** D7 and over. A steep sea, short for its height, goes up one colour, so the sea can turn red before the wind does.

The estimate is worked out from the wind the face has logged over up to 72 hours, counting only the time the wind has held roughly the same direction (within 30°), and from the **Sea fetch** setting: the stretch of open water the wind blows over. Use 200 nm (the default) for most coastal and offshore sailing, 500 or 1,000 nm in the open ocean, 50 or 100 nm in a sound or narrow sea.

Wind, steady	200 nm, after 24 h	1,000 nm, after 72 h
20 kn	D5 (about 3.1 m)	D5 (about 3.3 m)
30 kn	D6 (about 5.0 m)	D7 (about 8.4 m)

- It estimates only the wind sea, not swell from weather elsewhere, which offshore can be the bigger part of the sea.
- It needs the wind reports, which come through your phone: when the wind triangle is gone, so is the estimate. A logged observation still shows.
- Like the tides, it is a guide for planning, not a measurement.

Your own observations (WeatherGlass Log)

Crew copies only: WeatherGlass Log and these features are not in the store version.

Offshore, without the phone, there is no forecast wind. **WeatherGlass Log** is a small companion app that takes your own observation and hands it to the face. Open it, answer the questions, and it closes. Near shore, with the phone in range, the wind and air temperature questions open on Garmin's current weather, marked **Garmin**: press START to accept, or scroll to change. Pressing straight through a screen never records a value nobody chose.

1. **Wind from**: 32 points, or Calm.
2. **Wind, knots**: 5-knot steps.
3. **Sea state**: Douglas scale 0 to 9, with the wave heights shown.
4. **Swell from**: 16 points, opening on the wind direction; or Skip (which skips the next two).
5. **Swell period**: Short (under 8 s), Average (8 to 12 s) or Long (over 12 s).
6. **Swell height**: metres or feet (set in the Log menu).
7. **Cloud, eighths**: 0 to 8, or sky obscured.
8. **Most cloud to**: 8 points, when the sky is partly covered.
9. **Cloud type**: Cb (squalls), TCu (showers building), Cu (fair), St (low grey layer), Ns (rain cloud, front), Ci (high cloud thickening, front coming).
10. **Visibility**: 0 (under 50 m) to 9 (over 50 km); opens on the last one you logged this leg, else on Skip.
11. **Air temperature**: whole degrees, or Skip. °C or °F follows the watch (System, Units, Temperature).
12. **Sea temperature**: the same way; opens on the air temperature minus your **Air-sea diff** setting, else on the last one you logged, else on Skip.
13. **Course steered**: 10-degree steps, opening on the GPS course; or Skip when not under way.

For the next 3 hours the face uses it:

On the face	What changes
Wind triangle	Your wind, with a cream border. No border: Garmin's forecast wind.
Forecast line	Your sea state, for example D5 obs, in the same colours as the estimate.
Moon spot (no port set)	A cloud symbol: a circle filled in eighths, a pointer toward the main cloud, and the type beside it.
Beside the pressure	Your air temperature in cream, for example 18°C.

The Log menu. The first item when the app opens is **Log** (it reads **Send N to log** when observations are waiting). It opens:

- **Send N unsent**: the observations not yet scanned to the iPad.
- **Send all**: every observation kept this leg, to refill a lost or reset iPad.
- **Prompts**: watch prompts on or off.
- **Swell height**: metres or feet.
- **Air-sea diff**: how much colder the sea usually is than the air, -10 to +10 (positive = sea colder).
- **End leg**: clears every observation, after a confirmation. The only delete.

Into bashnav. Each observation is kept on the watch, up to 300 (about a leg at 3-hourly), with your position, course and speed from GPS (the face also takes the position, to follow the boat), the pressure from the barometer, and your heart rate and Body Battery, as bashnav log records. Sending shows them as QR codes, one after another, about one a second, for bashnav's scanner on the iPad to collect in one pass; an observation that does not fit in one code goes as two (the label shows 1/2, 2/2). UP and DOWN step by hand and pause. SELECT asks to mark them sent; they stay on the watch until End leg.

Prompts. Switch **Prompts** on in the Log menu when you sail; it is off by default. While you are up (Do Not Disturb / sleep mode off), the watch offers to open the app at the start of your watch and then every 3 hours after your last observation. Turn Do Not Disturb on when you go off watch and the prompts stop. The face also shows **OBS DUE** under the moon or port spot when one is due.

Your wind also feeds the sea-state estimate. To open the app quickly, map it to a hot key (Settings, System, Hot Keys) or add it to the controls menu, if your watch allows.

Sun, moon and tides

Each event is a small mark on the dial edge at its time of day, read like the hour hand: a mark just past the 6 means about half past six. Each mark shows the next time that event happens. Where a mark meets the wind triangle, the mark is drawn on top. With **Steps and battery** set to Thin arc (the default), the sun and moon marks are drawn a quarter larger, since the rim has room.

Mark	Meaning
Yellow half-sun	Sunrise
Orange half-sun	Sunset
Bright silver moon	Moonrise, drawn in today's phase
Dim silver moon	Moonset, drawn in today's phase
Outline circle	Moonrise or moonset at new moon
Blue crest over a line	Next high water
Blue trough under a line	Next low water
Small ☉, yellow or orange	Compass bearing of sunrise or sunset (see Sun and moon compass)
Small ☽, bright or dim silver	Compass bearing of moonrise or moonset

Springs and neaps. The tide marks fill with green from the left as spring tides approach: almost empty at neaps, full on the spring day itself. The fill uses each port's own delay between new or full moon and springs, which is usually a day or two.

Sunrise and sunset come from the watch itself. Moon and tide marks need a position or a port (see the next section). Tides always come from a port: the nearest one when the face has a position, else the port you choose.

Choosing your port

The face follows the boat when it knows where it is, and uses your chosen port when it does not.

Where the position comes from. The newest of: the watch's GPS (during an activity, or its last saved fix), the position of the phone's weather report, and the GPS fix WeatherGlass Log sends with each observation. A position counts as current for 6 hours, which covers 3-hourly observations with one missed.

Port setting	With a current position	Without one
Auto	The position, in degrees and minutes on two lines, in green. Sun and moon for the position; tides from the nearest port with tide marks.	The moon phase, as for None.

Port setting	With a current position	Without one
A named port	The position, in amber (red in Rig for red): the port is set but not in use. Sun and moon for the position; tides from the nearest port.	The port's name. Moon and tides for that port.
None	The moon phase; sun, moon and compass for the position. No tide marks.	The moon phase; moon marks from the last fix the watch has, if any. No tide marks, no compass.

The nearest port is chosen from every port whose tide marks are reliable, however far away, and is checked again when the boat moves about a mile. Its name is not shown: the spot shows where you are, not which port the tides come from.

Pick **Auto** for sailing. Pick a named port when you want its tides with no position, for example ashore with the phone left behind; seeing its name is your reminder to change it when the boat moves on.

On the watch:

1. Hold the UP/MENU button, choose **Watch Face**, then **Customize** on WeatherGlass.
2. Choose **Port**, then **Auto**, or a group: **Favourites** or **Leg 1** to **Leg 8**.
3. Choose the port. **None** returns the moon phase to the centre.

In the Garmin Connect IQ phone app (store installs): Port is one list, starting with Auto, each port marked with its group, for example "L2: Cape Town".

Group	Ports
Favourites	Stamford, Newport RI, Portsmouth UK, Halifax, Bar Harbor, Portsmouth NH, Gloucester MA
Leg 1	Southampton, Plymouth, London, Southend, Liverpool, Hull, Holyhead, Brest, La Rochelle, Cascais, Portimao, Vilamoura, Puerto Sherry, Funchal, Salvador, Rio de Janeiro, Punta del Este
Leg 2	Cape Town, Durban
Leg 3	Fremantle, Albany, Geraldton, Port Louis, Mahe, Christmas Island
Leg 4	Sydney, Newcastle, Hobart, Brisbane, Gold Coast, Airlie Beach, Tauranga
Leg 5	Singapore, Batam, Subic Bay, Da Nang, Ha Long Bay, Sanya, Zhuhai, Hong Kong
Leg 6	Qingdao, Tongyeong, Shanghai, Yokohama, Naha, Honolulu, Victoria BC, Seattle, San Francisco, Santa Cruz
Leg 7	Panama City, Colon, Galapagos, Fort Lauderdale, Nassau, Havana, Port Antonio, St Thomas, Bermuda, Washington DC, New York
Leg 8	Sydney NS, St Johns NL, Horta, Cork, Derry, Oban, St Helier, Den Helder, IJmuiden

What the port name tells you (and, with a position, the tides from the nearest port):

- **Normal green name**: tide marks are a good guide.
- **Dimmed name**: tide marks are a rough guide, because the port's two daily tides are uneven (for example Seattle, San Francisco, Honolulu), or because of the Solent's double high water (Portsmouth UK, Southampton, where the first high water is marked).
- **No tide marks, moon only**: ports where the tide comes mostly once a day or is very small (for example Fremantle, St Thomas, Punta del Este), and ports up rivers or estuaries (London, Hull, Derry, Shanghai, Hong Kong).

Sun and moon compass

With a position (or a named port), the dial becomes a north-up compass card, in the same frame as the wind triangle: the triangle at 12 is north, and small E, S and W letters sit beside the 3, 6 and 9. A light cream **pointer hand** points at the sun's true bearing, with ☉ at its tip, while the sun is up.



Using it. Turn your wrist until the pointer hand points at the sun. The 12 triangle then points true north, the letters show east, south and west, and the wind triangle shows the true wind direction around you. No compass or magnetometer is needed: the bearing is worked out from the time and position.

At night (setting **Compass at night**): **Moon hand** (default) points the hand at the moon while it is up, with ☽ at its tip and in the moon marks' silver; **Hide** removes the hand and letters; **Letters only** keeps the letters.

Where the sun and moon rise and set (setting **Rise and set marks**):

- **Times and bearings** (default): the sun and moon marks stay at their clock times, and small ☉ and ☽ symbols sit at their compass bearings: yellow ☉ sunrise, orange ☉ sunset, bright ☽ moonrise, dim ☽ moonset. With the card lined up, each symbol points where on the horizon the sun or moon will rise or set.
- **Bearings**: the marks move to their bearings, and their times are not shown.
- **Bearings by day**: bearings while the sun is up, times at night.

Tides always stay at their times, since a tide has no bearing.

Two more choices:

- **Compass letters**: **Beside** (default) puts the letters just outside the 3, 6 and 9; **Replace** draws E, S and W in place of those numerals.
- **Sun hand**: **Full** (default) runs from the centre; **Outer** starts at the edge of the time window, so it never crosses the time.

The compass is not shown on the sleeping screen. With no position and no port, there is no hand and no letters.

Time, heart rate, steps and battery

- **Local time** is the large number in the window at the bottom, in 24-hour or 12-hour form (setting).
- **Day, date and UTC** sit on one line above it, for example TUE 3 15:00 Z. UTC is always 24-hour: the time logs, sights and bulletins use.
- **Heart rate** shows the current beats per minute and a 1-hour trace. The number and the trace change colour with your training zones from Garmin Connect: green up to zone 3, amber in zone 4, red in zone 5.
- **Steps** fill the left side of the rim, upward from 7 o'clock, full at 20,000, with a mark at 10,000.
- **Battery** fills the right side of the rim, upward from 5 o'clock; below 20 % it turns red.

The **Steps and battery** setting picks how they are drawn:

Choice	Looks like
Thin arc (default)	A thin dark-green arc on the rim, the unfilled part faint; no minute ticks, only the hour marks. Sun and moon marks are drawn larger.
Long ticks	The minute ticks carry steps and battery: lit ticks green, longer and thicker; unlit ticks faint.
Ticks	The same, lit ticks the normal length.
Arcs	Segmented arcs inside the minute ticks: one segment per 1,000 steps or 10 % of battery.

Hour marks: **Fine** (default) are long thin red marks at the rim; **Bold** are the earlier short, thick marks.

Pub reminder

Switch it on and, every day at your chosen time, a pint appears on the face for 10 minutes with PUB and the name of your beer. While it shows, the line above gives the local time beside UTC, so the time is never lost. Suggested by Els, first mate of the Level 1 training crew.

To set it up on the watch:

1. Hold UP/MENU, choose **Watch Face**, then **Customize** on WeatherGlass.
2. Turn **Pub reminder** on.
3. Choose **Pub time**, pick the hour, then :00, :15, :30 or :45 (default 17:00).
4. Choose **Beer**: Murphy's (default), Guinness, Beamish, London Pride, Doom Bar, Lager or Cider. The pint is drawn as a stout, ale, lager or cider to match.
5. Optional: **Pint position** switches between **Time window** (default, large) and **Moon spot** (smaller, leaving the time window as normal).

In the phone app you can also type any beer name of up to 9 letters; it replaces the list name on the face.

A watch face cannot vibrate or wake the screen, so the pint is waiting whenever you look at the watch during those 10 minutes; it does not buzz.

Frans mode

A crew joke, after Frans's greeting to his mates. Switch it on and, once a day, at a random moment, a drawn hand with the middle finger raised appears on the face for 2 minutes. No words.

It only appears when you are up and calm: between an hour after your usual wake time and an hour before bed (from your Garmin profile), with your stress reading under 50 and your heart rate out of the workout

zones. It shows the first time you look at the watch after its moment, in the same place as the pub reminder's pint (see **Pint position**).

In the version from the Connect IQ store, the hand gives a palm-out V sign instead.

To switch it on: hold UP/MENU, choose **Watch Face**, then **Customize**, and turn on **Frans mode**. It is off by default.

Settings reference

Every setting is on the watch under UP/MENU, **Watch Face**, **Customize**; store installs also have them in the Garmin Connect IQ phone app.

Setting	Choices	Default
Seconds hand	On / Off	On
24-hour time	On / Off	On
All 12 numerals	On / Off (off shows 3, 6 and 9 only)	Off
Pressure in inHg	On / Off (off shows hPa)	Off
Colour	Red + amber + green / Rig for red	Red + amber + green
Hemisphere	Auto / North / South (also sets the side of the wind barbs)	Auto
Port	Auto, None, or a port from Favourites or Leg 1 to 8	None
Compass at night	Moon hand / Hide / Letters only	Moon hand
Rise and set marks	Times and bearings / Bearings / Bearings by day	Times and bearings
Compass letters	Beside / Replace	Beside
Sun hand	Full / Outer	Full
Steps and battery	Thin arc / Long ticks / Ticks / Arcs	Thin arc
Hour marks	Fine / Bold	Fine
Wind barbs	On / Off	Off
Sea fetch	50, 100, 200, 500 or 1,000 nm	200 nm
Sea height in feet	No longer changes the face (the sea state is a Douglas number)	Off
Pub reminder	On / Off	Off
Pub time	Any quarter hour	17:00
Pint position	Time window / Moon spot (also used by Frans mode)	Time window
Beer	Murphy's, Guinness, Beamish, London Pride, Doom Bar, Lager, Cider	Murphy's
Beer name (phone only)	Up to 9 letters, replaces the list name	Blank
Frans mode	On / Off	Off

Night use and the sleeping screen

The standard colours are chosen to be read at night with the watch's red-shift mode off: red hands and numerals, amber for structure, green for readings, and the one blue kept to thin tide lines. For full dark

adaptation on deck, set **Colour** to **Rig for red**, which turns everything red.

When the screen sleeps (always-on display), the face shows only the hands, four markers and the digital time, dimmed and shifted a few pixels each minute to protect the screen. Edge marks, the compass, the barometer and the pub reminder return as soon as you raise your wrist.

Supported watches and installing

WeatherGlass runs on Garmin's larger AMOLED round watches:

- quatix 7 Pro, quatix 8 (47 and 51 mm)
- epix Pro (Gen 2) 47 and 51 mm
- fēnix 8 AMOLED 47 and 51 mm, tactix 8 (47 and 51 mm)
- fēnix 9 47 mm, fēnix 9 Pro 47 and 51 mm
- MARQ (Gen 3)

Also supported, on the older always-on MIP screen: the fēnix 7X, 7X Solar, 7X Sapphire Solar and 7X Pro, tactix 7 and quatix 7X Solar. WeatherGlass is tested on both screen types: AMOLED on a quatix 7 Pro, and MIP on a fēnix 7X Sapphire Solar. Crew copies are also built for the **quatix 7** (and fēnix 7) on its 260 px MIP screen, and the **quatix 7 Sapphire AMOLED** (and epix Gen 2): they join the store version once checked on a real watch. Other MIP watches (such as the fēnix 8 Solar) and the smaller AMOLED models are not supported yet. If yours is one of them, say so; it helps decide what comes next.

Temperature units on the face and in WeatherGlass Log follow the watch: hold UP/MENU, System, Units, Temperature.

Installing. WeatherGlass is free on the Garmin Connect IQ store: https://apps.garmin.com/en-US/apps/247dedb7-9de8-463f-ab73-33eee20b7695. Crew copies come from Larry as a .prg file: plug the watch into a computer (on a Mac, with the free OpenMTP app), copy the file into GARMIN, then APPS, and unplug. Then hold UP/MENU on the watch, choose **Watch Face**, and select WeatherGlass.

Good to know

- **Tides are a planning guide, not for pilotage.** High and low water are typically within about half an hour of the published tables, closest at neaps. Weather, river flow and local effects move real tides; use proper tide tables for passage planning.
- **Barometer ashore.** The face records the watch's sea-level pressure every 5 minutes, which the watch corrects for altitude, so a hike or a drive over hills does not show as weather. After a very long day out (over about 2½ hours away from the face) with a change of altitude, the trace turns grey and the forecast line reads ALTITUDE CHANGED until it has fresh readings: grey means unknown, not calm. Keep the watch's barometer Sensor Mode on Auto (the default). At sea none of this arises.
- **Position and tides.** With a position less than 6 hours old, the sun, moon and compass use it, and tides come from the nearest port; otherwise everything uses the port you chose. Offshore with no phone and no activity recording, WeatherGlass Log's observations keep the position fresh.
- **Wind** needs the phone: it is Garmin's forecast wind, refreshed through Garmin Connect.
- **Phone settings** appear only when the face is installed from the Connect IQ store; a test copy is set up on the watch.

Credits

- Design: Larry Sherman, with the Clipper Race Training shipmates.
- Forecast: the Zambretti forecaster, after beteljuice's formulation.

- Tidal constants: NOAA CO-OPS station data (US ports) and NASA's GOT5.5 global ocean tide model.
- Pressure tendency terms: the Met Office shipping forecast.